



When Fishery Rhymes with Poverty: A First Step Beyond the Old Paradigm on Poverty in Small-Scale Fisheries

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Summary. — In this paper, we first look retrospectively at the perceptions embraced by academics, international agencies and practitioners of the relation between fisheries and poverty in developing countries and we try to identify the underlying paradigms which have structured these perceptions. The review reveals how the debate has focused on the economic (low income) and biological (overexploitation) aspects of the problem. We then revisit these perceptions in the light of the recent conclusions drawn from other sectors and in particular from the new “consensus” on poverty proposed by the international community. Incorporation of the recent research on poverty helps to show how socio-institutional mechanisms governing people’s access to fisheries resources—rather than the resources themselves—play such a critical role in vulnerability to poverty. Finally, a typology identifying four different categories of intrasectoral exclusion mechanisms is developed and illustrated through empirical studies derived from African and Asian fisheries.

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1. INTRODUCTION

Throughout the world, but especially in Asia and Africa, small-scale fishing communities present all the characteristics that should have caught the full attention of both experts in development and academics interested in poverty issues. First, because the degree of poverty and deprivation that characterize small-scale fishing individual households and communities have long been recognized and reported. For instance, a few decades ago FAO emphasized that “the people engaged in these activities and their families continue, with few exceptions, to live at the margin of subsistence and human dignity” (FAO, 1974, quoted in Copes, 1989, p. 6). Second, because many small-scale fisheries communities (especially in West Africa and Asia) are characterised by a constellation of local organizations, forms of collective action, processes, and interactions among individuals groups, and institutions which constitute the main themes of research in domains such as social capital (Pretty & Ward, 2001), empowerment (Botchway, 2001; Friedmann, 1992), poverty and environment nexus (Cavendish, 2000; Leach, Mearns, & Scoones, 1999; Rear-

don & Vosti, 1995), vulnerability (Chambers, 1989), local institutions (Agrawal & Gibson, 1999) and participation and human capabilities (Bass & Rouse, 1997; Beck & Nesmith, 2001).

Yet, there is almost a complete absence of references to fisheries case studies in the current literature on poverty. For illustration, in the recent 2000/2001 World Development Report *Attacking poverty*, the keyword “fish” (including fish, fishing, fishers, fishermen,

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fisherwomen, etc.) appears only three times in 266 pages.¹ The absence of references on fisheries² also exists in academic research. For example, the thorough review made very recently by Beck and Nesmith (2001) in India and West Africa on the role of common property resources in poor people's livelihoods includes only one fisheries case study (an inland fishery community in Nigeria—Olomola, 1998) among the 35 case studies reviewed (18 in India and 17 in West Africa).

Why is a domain so "rich" in empirical evidence of poverty and examples of social organizations and regulations, collective action, and institutional arrangements so neglected in the current debate on poverty?

The objective of the present article is to attempt to answer this question. For this, the paper is articulated in a two-step analysis. First, it demonstrates from a review of the literature how empirical studies addressing the question of poverty in fisheries have been largely influenced by what can be termed the "old paradigm" on poverty in small-scale fisheries. In brief, this old paradigm is the perception that poverty in fisheries is mainly related to the natural factors (fishing resource) and its associated exploitation level. This is an illustration, in the fishery context, of the Malthusian perception of poverty (the lack of resources or their overexploitation due to population growth leads to poverty and famine). Second, the analysis will be broadened by placing the lessons derived from this review in the wider context of recent conclusions on poverty proposed by the international development community. In particular, the results outlined in the fisheries case are "revisited" through a new approach focusing on the socio-institutional dimension of poverty.

By so doing, the article emphasizes two points. First, the fact that despite the indisputable existence of cases where community poverty is induced by, or directly related to the overexploitation or depletion of the resource, focus on the biological dimension of the issue (overexploitation of the resource) has contributed to overlooking other elements which call for a wider analysis of poverty in fisheries, in particular the institutional dynamics of this poverty. Second, the fact that fisheries represent an almost "bottomless pool" of empirical examples of different types of socio-institutional failures leading to, or maintaining, poverty in rural communities and therefore that the integration of these examples in the current research would no doubt contribute to a better

understanding of the various mechanisms related to poverty in rural areas.

In this article, "socio-institutional mechanisms" are understood as the set of rules, norms, conventions, institutional arrangements, forms of co-ordination or social (class) organization, and decision-making processes through which the social actors (individuals and collectives) interact to gain and maintain their command (access, use and control) over the resources and the benefits extracted from them. As such, the socio-institutional dimensions of fisheries activities include the rules and regulations (formal and informal) governing the access to the fishing resources or to the subsequent benefits, but also the political (power-related), economic, social, and cultural resources and constraints which condition this access and its control. It is within these social and institutional dimensions—and this is the thesis of this article—masked or underestimated by the old paradigm that lie the more important causes of poverty in fisheries. Beyond the debates on the relationships between environment and poverty (the so-called poverty–environment nexus), this essay proposes to demonstrate that the social and institutional mechanisms which take place within and around the fisheries play a very important role in the maintenance, alleviation, or aggravation of poverty in fisheries-dependent communities.

2. METHODOLOGY

The data used in this article were gathered through an extensive literature review. For the first section (description and analysis of the old paradigm and the perception of poverty in fisheries), published articles, discussion papers and international institution reports (e.g., FAO, World Bank) were examined. They concern small-scale fisheries from all over the world. For the second section (analysis of the socio-institutional dimension of poverty in fisheries) although worldwide case studies were initially examined (through both published and "gray" literature), it appeared that the very few empirical studies which did not simply report poverty but also attempted to analyze its socio-institutional mechanisms are essentially from Asia (India, Indonesia, Bangladesh, Sri Lanka) and a few from Africa (Cameroon, Congo, Nigeria). This initial account was enlarged by the author's recent experience in Nigeria, Cameroon, and Chad, as part of an interna-

tional research project on Lake Chad Basin fisheries (Neiland & Béné, 2002). Altogether, the case studies used to cover both coastal and inland fisheries, from Asia and Africa. Because the objective of the article is to make a precise and exact report on the 'state-of-the-art' on research on poverty in fishery, direct quotations are used as often as possible to illustrate objectively this state-of-the-art.

3. POVERTY IN FISHERIES: REVIEW OF THE LITERATURE

One possible explanation for the absence of fisheries case studies in the debate/research on poverty could be the lack of field experience in that particular domain from the analysts and experts involved in the current research on poverty.³ This lack of specific experience in the fisheries domain could however potentially have been eased by the large number of empirical studies that mention the occurrence of poverty in small-scale fisheries communities. Even a brief review of the literature would indeed uncover numerous reports, papers and case studies which emphasize the apparently severe deprivation characterising fisheries-dependent communities. The amount of available literature on fisheries cannot therefore be invoked to explain the poor input of fisheries research to the debate on poverty. It seems that the explanation lies elsewhere, probably in the nature of this scientific production and in the hypotheses that this literature proposes in general to explain the cause(s) and origin(s) of poverty in small-scale fisheries.

(a) *Fishermen: the "poorest of the poor"*

There are in the literature two contrasting interpretations of the relationship between poverty and fisheries. The first relates to the view that "they are poor because they are fishermen." This is the "endemic poverty" perception of poverty in fisheries in which it is felt that fishermen, whatever they try to do, will remain poor. The apparent correlation between fisheries and poverty has been, highlighted by numerous observers (see Table 1 for some examples). Bailey (1988) perfectly summarizes this commonly stated "reality" by noting that fishing communities are usually presented in the literature as "the poorest of the poor" (p. 36).⁴ This perception, which is still widely held among decision-makers and experts from in-

ternational agencies⁵ is based on two different logical arguments which, although derived initially from opposite assumptions, eventually join to suggest the same conclusion, namely that "fishery rhymes with poverty."

(i) *The conventional wisdom and the endogenous origin of poverty in fisheries*

According to the first of these two approaches, poverty in fisheries is related to the (low) level of the natural resources and its origins/causes are entrenched within the fisheries sector itself (endogenous origins). This approach is what Copes has called the "conventional wisdom on poverty in fisheries" (Copes, 1989, p. 10). This "conventional wisdom" is rooted in Gordon's seminal (1954) paper on fisheries economics and the open-access nature of fisheries. Gordon (who had been commissioned by the Canadian government to explain the persistent problem of low incomes among Canadian fishermen⁶) concluded that it was the rent dissipation (induced by what he termed the "common property nature" (p. 134) of the fisheries) that leads to the low income of fishermen. This idea was then reframed by Hardin in his famous (1968) paper "The tragedy of the commons." Interestingly, although Hardin's article has since been over-used to emphasize the biological outcome of the tragedy (the overexploitation of the resources), the author (a biologist) had also largely predicted the economic dimension of it:

Therein is the tragedy [of the "commons"]. (. . .) *Ruin* is the destination toward which all men rush, each pursuing his own interest in a society that believes in the freedom of the commons. Freedom in a common brings *ruin* to all (Hardin, 1968, p. 1244) (emphasis added).

Acknowledging Hardin and Gordon's confusion between common property (*res communes*) and open access (*res nullius*),⁷ the logic of the conventional wisdom is straightforward and reads as follows: the open-access nature of the fisheries allows more and more people to enter the fishing sector, which leads to the economic (and possibly biological) overexploitation of the resources—thus the Malthusian dimension of poverty,⁸ dilapidation of the economic rent, and finally impoverishment of the fishing community.⁹ Table 2 lists some of the numerous quotes found in the literature which support this perception.

Although it will be argued below that it is the overemphasis of the economic (low income) and biological (overfishing of the resource)

Table 1. *The apparent equation between poverty and fisheries*

Reference (chronological order)	Quotation
FAO (1974) (quoted in Copes, 1989, p. 6)	"The people engaged in these activities and their families continue, with few exceptions, to live at the margin of subsistence and human dignity"
Smith (1979, p. 2)	"The general condition of poverty characteristic of traditional fishing communities around the world have increasingly drawn the attention of governments and other change agents in recent years"
Panayotou (1982, p. 1)	"The fundamental problem of small-scale fishermen around the developing world is their persisting absolute and relative poverty"
Bailey, Cycon, and Morris (1986, p. 1270)	"Yet, despite (and sometimes because of) significant investment by international donor agencies and the governments of developing nations, small-scale fishermen remain among the poorest of the poor within developing nations"
Juntarashote and Chuenpakdee (1991, p. 556)	"It has been commonly understood that as a concept the small-scale fishery is the fishery in need of assistance from government, as fishermen who engage in this fishery are generally poor"
Copes (1989, p. 2)	"In attempting to explain why many fishermen have low incomes, one needs to proceed from an understanding that fishing incomes indeed are often low. The literature abounds with allusion to the relative poverty of fishermen"
Wright (1990, p. 2)	"The cause and the nature of the poverty that persists in many fishing communities is important"
World Bank (1992, p. 2)	"The harvesting, processing and marketing of these resources is a source of income for over 100 million people, about 80% of whom are in the low income or poverty group"
Cunningham (1993, p. 2)	"The evidence that fishermen's incomes are generally low is overwhelming"
Chowdhury (1993, p. 8) (quoted in Kremer, 1994, p. 5)	"They [the fishermen] live sub-human lives and are the most deprived and poorest class"

aspects that led to the narrow interpretation of the phenomenon of poverty in fishing communities, it must also be acknowledged that resource limitations do exist, overexploitation does take place and that

overexploitation has a particularly serious effect on employment opportunities and standards of living for small-scale fishing households, which represent the vast majority of the people who depend on fishing for their livelihood (Bailey & Jentoft, 1990, p. 334).

There is indeed much evidence throughout the world of small-scale fisheries where the resources have been (or are being) overexploited—either by the communities themselves or by "outsiders", e.g., industrial fleets—and where this overexploitation greatly impinges

upon fishermen's livelihoods. Kurien, for instance, in his review of the development of Kerala (South India) coastal fisheries showed how the introduction of modern trawlers in the 1970s led to the overexploitation of the local stocks and how this negatively affected the local population (in terms of incomes, but also food supply). He reported:

As productivity has declined, the major economic brunt of overfishing has been born by fishworkers and their families. (...) In the case of the artisanal fishermen, ... productivity registered a 50 percent decline between 1974 and 1982, and real per capita incomes dropped from 850 rupees to 420. (...) Meanwhile the adverse nutritional impact has been borne by the local consumer—particularly the poorer among them, [including the local fishermen themselves]. (...) Per capita availability of locally consumed fish have decreased

Table 2. *Poverty in fisheries and the tragedy of the "open access"*

Reference (chronological order)	Quotation
Gordon (1954, p. 134) ^a	"That the plight of fishermen and the inefficiency of fisheries production stems from the common property nature of the resources of the sea is further corroborated by the fact that one finds similar patterns of exploitation and similar problems in other cases of open resources"
Sinclair (1977, pp. 5–6) ^a	"A considerable amount has been written about the economic waste that is generated when fisheries are managed under open access conditions. . . Standard analysis shows that in fisheries operating under these conditions all potential economic rent is dissipated and that the average fisherman . . . will earn less than he would in other occupations requiring similar skills"
Pearse (1982, pp. 75–76) ^a	"All of these effects—stock depletion, poor economic performance and instability—result from treating the resource (the fish) as common property until they are caught. . ."
Panayotou (1982, p. 29)	"Thus we may conclude that open access, (. . .) may lead not only to the dissipation of economic surplus for the society at large but also to the impoverishment of the fishermen themselves"
Steedman (1985, p. 9)	" . . . small-scale fishermen occupy a position near the bottom of the income scale. Livelihoods are meagre for a variety of socio-economic and environmental reasons . . . Among these is the open access of the resource itself, which leads to overfishing in coastal and inland waters and results in dwindling stock, poor catch, and a perpetual cycle of poverty among fishermen"
Heaps and Helliwell (1985, p. 430) ^a	" . . . this open access equilibrium dissipates the wealth (or rent) . . . The result has been that excessive effort is used in the fishery, fish stocks may be dramatically reduced and fishermen tend to remain poor with incomes little more than their opportunity incomes"
Clark (1985, p. 1) ^a	"Gordon's theory of the common property fishery . . . not only explained the low income of fishermen, but also clarified in economic terms the so-called overfishing problem"
FAO (2000a, point 5)	"In fact, it has been often argued that the open access characteristic of many marine and inland fisheries is one of the important causes for widespread and persistent poverty in fisheries"

^a Also quoted in Wright (1990, Appendix A).

from around 19 kilogrammes in 1971–72 to around 9 kilogrammes in 1981–82 (Kurien, 1993, p. 7).

Through the "story" of the Kerala artisanal fisheries, Kurien perfectly demonstrates the mechanical and almost irreversible dynamics that link the overexploitation of the resources to the degradation of the living conditions of the dependent communities.¹⁰ This example is unfortunately just one among many empirical cases¹¹ that recall that fishery resources are, by nature, limited; that these resources are consequently subject to possible overexploitation; and that overexploitation can indeed be a major—if not *the* major—cause of impoverishment for fisheries-dependent communities. This risk of "overexploitation-induced pov-

erty" is therefore a reality that cannot be underestimated.

(ii) *The exogenous origin of poverty*

The second major school of thought which has attempted to address the issue of poverty in the fishery sector has based its argument on a different approach and in particular on the economic concept of "low opportunity incomes." The assumption here is that because alternative incomes (outside the fisheries sector) are usually low, this drives (or keeps) fishermen's incomes at low levels through mechanisms of labor transfer between sectors. Although it is more difficult to attribute this assumption to one single source, one key contribution is certainly Cunningham's 1993 Paper

"Fishermen's incomes and fisheries management".¹² Cunningham argues that, assuming an open-access fishery, perfect labor mobility, and a standard two-sector model (a fishery and a nonfishery sector), it is easy to show¹³ that even if the fishery sector generates economic surplus in the short term, there will be pressure to enter the sector, resulting in a long-term wage re-equilibrium between the two sectors.

Given that—particularly in developing countries—small-scale fisheries are usually located in rural, remote areas, with very few alternative employment opportunities, this argument has been, and is still widely used to explain poverty in fisheries. Smith (1979, p. 10) and Panayotou (1982, p. 18) had previously highlighted this lack of alternative employment as one of the key factors contributing to a low standard of living in small-scale fisheries. Panayotou, for instance, concluded: "most often, the reason behind the absolute, if not relative poverty of small-scale fishermen is the lack of sufficiently attractive alternatives" (Panayotou, 1982, p. 29). Within this approach, and in contrast to the conventional wisdom, the roots of poverty "stem" from outside the fisheries sector. The low income of the fishermen (assumed to be equivalent to poverty) is not related to the exploitation level of the resource or to the dissipation of the rent (Copes, 1989, pp. 6 & 10; Cunningham, 1993, p. 6), but to the economic situation outside the fishery.

(iii) *The resulting perception: Fishery = poverty*

Although both the Gordon and Cunningham studies (based on standard neoclassical economics, e.g. assuming perfect market mechanisms) were looking at the origin of *low income* in fisheries, it is interesting to note that the rationales on which these studies were based have been diverted from their initial restricted application domain and used to explain the occurrence of *poverty* in fisheries, in particular in developing countries' small-scale fisheries (where, however, the assumption of perfect markets may be more disputable). It is also interesting to note that although they do not recognize the same origin of poverty in fishing communities (the former assuming an inside-sector poverty tied to the exploitation level, the latter assuming an outside-sector poverty independent to the exploitation level), they have very often been treated as a combined, single phenomenon where the internal mechanism of the "tragedy of the open-access" is presented in combination with the external, low income

earning opportunities to "explain" poverty in fisheries as a whole. For instance, Bailey and Jentoft stated (1990, p. 335):

Unlike agricultural land ... fisheries resources generally are open access resources where no boundaries exist and no restrictions are imposed on who may become a fisher or how the resource may be exploited ... this dismal conclusion is the essence of Hardin's tragedy of the commons and reflects what has occurred in many third world fisheries.

Clearly Bailey and Jentoft (1990) refer here to the "inside" mechanism leading to poverty. The authors then conclude (p. 337):

Labour mobility serves to indicate the close connection between fisheries and other sectors of the national economy and suggests that problems of fisheries development cannot be solved by reference to the fisheries sector alone. (...) The only way out of this conundrum is to improve economic opportunities outside the fisheries.

Likewise, FAO stated recently that "Unconstrained labor mobility and open access [inside mechanisms] ... presuppose that poverty in fisheries will continue to persist as long as there is poverty elsewhere in the country [outside mechanisms]" (FAO, 2000a, point 5).

Other authors explicitly recognize the added effect (combination) of the two mechanisms. Smith (1979, p. 9), for instance, wrote:

The lack of alternative income opportunities in the rural sector [outside cause] which intensifies fishing effort *then couples with* limited resources [inside cause] to reduce income levels (emphasis added).

The thesis of this article is that this mixing between the various natures/structures of poverty in fisheries contributed to the persistence of an overall impression of strong causal relationship between fisheries and poverty, and failed to facilitate the emergence of an appropriate research framework. A consequence (or illustration) of this lack of analytical framework is the repeated amalgam made between fishermen's low income and poverty. Although not specific to fisheries literature,¹⁴ this confusion in a sector where the subsistence dimension of the activity is often very high, especially in developing countries, prevented the formulation of pertinent questions about poverty in fisheries. The general view was: "They are poor because they are fishermen." This perception constitutes the first element of the old paradigm.

(b) *The distributive dimension of fisheries: The "last resort activity" argument*

The second major idea commonly conveyed in the literature about fisheries and poverty refers to the distributive role of fisheries, namely the fact that fisheries (because of their "open-access" nature) offer poorest people a livelihood through fishing activities. This is the perception of small-scale fisheries as an employer of "last resort" or as "safety valve" for the poor. FAO, for instance, recently recalled this perception of safety valve mechanism:

Some have argued that open access (or quasi open access) to inshore and inland fisheries is desirable because it serves as an insurance and/or safety mechanism against shocks for poor people who have lost permanently or temporarily their means of survival on other economic sectors (or regions) (FAO, 2000a, point 24).

The review of the literature emphasizes the consensus among academics and institutions about the occurrence of this mechanism in developing countries, as illustrated in Table 3. This "last resort" dimension of fisheries is obviously of great importance and relevance to the poor, since the latter are generally those with limited access to land and/or other resources. In that respect, a dimension of this "last resort" is comparable to the role that other common pool resources (CPRs) are recognized to play in poor's livelihoods. Beck and Nesmith (2001), for instance, provide evidence of the importance of CPRs such as forests, rangeland, bushland, fallow fields, inland waterways, wetlands, seasonal ponds, etc., for the poor in India and Africa. As a CPR, fisheries play therefore an important role for poor people. But, what the "last resort" hypothesis tends to assume or even to assert is that fisheries are not merely one among several CPRs but the very last one, i.e., the one toward which the poorest turn when they are denied access to the other CPRs. This "last resort" capacity obviously remains to be further investigated and some empirical studies presented below challenge this view.

The "open-access" nature of the resource appears in the present case as the key mechanism which permits this safety valve to function and allows people to enter the fisheries when their access to other activities or resources is (economically or institutionally) limited or impeded. It is therefore interesting to note that

within this approach, the "open-access" nature of the fisheries seems to be a positive factor—in total contrast to its perception under the conventional wisdom. If the poverty of households involved in fishing activities is still acknowledged, it has, in this case, a totally different character. As Dunn (1989, p. 4) emphasized:

Many communities of fishermen are poor but it should be realized that they are not necessarily poor because their livelihood is fishing. They are often already poor and landless individuals who are able to subsist by fishing.

In other words, under the "safety valve" approach the adage "They are poor because they are fishermen" might be reversed to read: "They are fishermen because they are poor (and landless)." This point has already been made by several authors. McKenzie, for instance remarks that "the received image of the poor fishermen is to be stood on its head—he is a fishermen because he is poor, not the way around" (McKenzie, 1979, p. 816 footnote 5, quoted in Cunningham, 1993, p. 2).

(c) *The two pillars of the old paradigm*

These two perceptions ("poor because fishermen" and "fishermen because poor") are in fact the two sides of the same coin and they represent the two complementary pillars of the old paradigm. Representing these two pillars together with the "low opportunity" model allows us to sketch the "global picture" synthesizing the way the relationship between small-scale fisheries and poverty has been "conceptualized" so far in the literature (Figure 1). The figure illustrates how the various scenarios complement and reinforce each other to maintain an overwhelming impression that fishermen are members of low-status, marginalized households, and how this impression eventually led to the equation "fisheries = poverty." Underpinning this equation is the circular and self-reinforcing logic of the "old paradigm."

4. TOWARD A NEW UNDERSTANDING OF POVERTY IN SMALL-SCALE FISHERIES

The objective of this section is to place the fisheries sector within the wider perspective of the current research on rural poverty. First, the

Table 3. *Fisheries as the "last resort for the poorest"*^a

Reference (chronological order)	Quotation
Panayotou (1980, p. 145)	"Open access fisheries (as well as other natural resources of similar status) often serve as safety valves for population pressures and as source of subsistence and cash income for those who by virtue of their poverty have little or no access to other sources of income"
Panayotou (1982, p. 30) (see also his footnote 32 p. 27)	"There is considerable evidence that coastal fishing is an economic activity of last resort. For example Pollnac and Sutinen (1979) report that in East Africa "fishing is viewed as the employment of last resort ... people fish when farming is not feasible" ... Yet another example is found in Cordell (1973) who reports that in Northeast Brazil labourers released from coconut plantations took up canoe fishing for lack of better alternatives ..."
Bailey and Jentoft (1990, pp. 337 & 341)	"The open access nature of fishery resources and the ease with which people can enter a fishery with limited experience or capital investment, means that there are few obstacle to seeking a livelihood at sea ... The role of fisheries as employment source of last resort underscores the importance of employment generation..." "...ocean fisheries function as a society valve for surplus labour ..."
FAO (1994, p. 7)	"Marine fisheries typically are the employer of last resort. Fishers who do not own the boats on which they work are reported to feel a general sense of hopelessness and despair regarding opportunities for upward mobility"
Machena and Kwaramba (1997, p. 245)	"The worsening economic climate in the country is also a strong contributory factor to the crisis. Fishing is often described as an 'activity of the last resort'. Entry into the gill-net fishery is easy compared to other economic ventures because of the low capital input requirements"
Townsley (1998, p. 142)	"Indeed, aquatic resources in many part of the world constitute a resource of 'last resort'. Open access fisheries and other forms of aquatic resource use are fall-back options for the rural poor when loss of land or failure in access to other rural activities threaten their livelihoods"
Paynes (2000, p. 1)	"[Fishery] is often one of the few livelihoods open to the landness and often becomes the default livelihood"
FAO (2000a, point 5)	"Small scale and subsistence fisheries have often been denoted as 'employers of the last resort' implying that people enter open access fisheries when they are unable to make a living in the other sectors"

^a Are quoted in this table only references which use explicitly the terms "last resort" or "safety valve," not those which mention or describe this mechanism but do not use the two key terms (for those, see for instance, van Zalinge, Thuok, & Seang Tana, 1998 or Degen & Thuok, 1998 for the rural populations in Cambodian floodplains after Pol Pot era, Beck & Ghosh, 2000 for the populations living on the Bank of the river Matla (India), Gutkind, 1989 for the Fante (Ghana) and Jul-Larsen, 1994 for the Xwala (Benin) communities during the development of the West African coastal fisheries in the 1950s and 1960s, Dia, 1998 for the cephalopod fisheries in Mauritania in the 1990s, etc.).

main themes that have characterized the debate and research on poverty over the last 30 years

will be examined briefly. For a more detailed analysis, please refer, for example to Shaffer

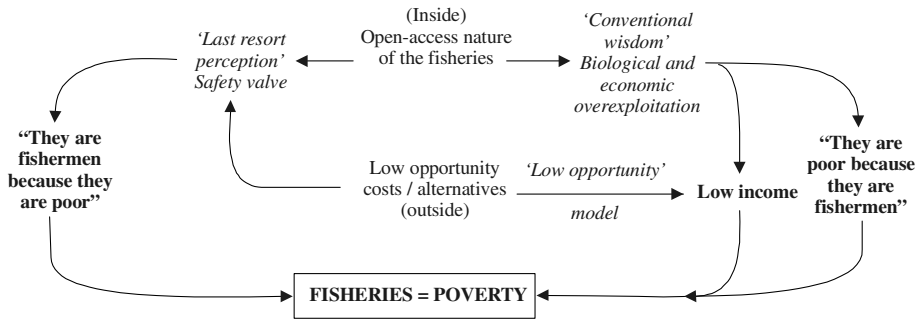


Figure 1. The two pillars ("they are fishermen because they are poor" and "they are poor because they are fishermen") forming the circular logic of the old paradigm and leading to the 'self-contained' equation "fisheries = poverty".

(1996), Baulch (1996), Lipton (1997), and Maxwell (2001).

(a) *From the concept of income-poverty to the consensus of multidimensional nature of poverty*

In the 1960s one of the most predominant criterion in development economics was income (reflected in macroeconomic indicators such as Gross National Product per head), and poverty was "naturally" assessed against income or consumption criteria.¹⁵ The underpinning rationale for this choice is straightforward and appears at first sight perfectly justified. Food is the most fundamental dimension of poverty and meeting basic food needs is the first priority of any individual.

Poverty and hunger, however—although closely associated—are not synonymous and poverty cannot be simply defined in terms of lacking access to sufficient food. In that context, the measurement of poverty through the level of income or consumption is not totally satisfactory. This limitation of the income-poverty model gave rise in the 1970s to the development of the "basic needs" model pioneered by the International Labor Organization (ILO) and UN Research Institute on Social Development (UNRISD). There was a new recognition that poverty was not just a failure to meet minimum nutrition or subsistence levels, but included a more general deprivation of the material requirements to meet minimally acceptable human needs such as health and education, clean water and other services required to sustain livelihoods. This basic needs model was the premise of a multidimensional definition of poverty which, later,

led to the human development model of the UNDP.

The 1980s marked an even more drastic redefining of the concept of poverty. Undoubtedly, a key element in this new in-depth treatment of poverty was the contribution of Amartya Sen. In his seminal 1981 book *Poverty and famine* Sen observed that famine can occur amid plenty and therefore that the absolute lack of resources is only one of the number of reasons why people lack access to the resources they need for sustaining their livelihood. Central to Sen's thesis was the concept of food entitlement. Sen argued that if people can starve in the midst of food surplus it is due to what he termed "(food) entitlement failure," i.e., a collapse in their means of command over food, governed by a range of social, economic, cultural, and political factors. This view brought a dramatic change in the way famine was perceived. Previously seen as a failure of production (a Malthusian crisis), famine was thereafter seen as a breakdown in the rights of access to food.

This idea of the failure of entitlement strongly activated the debate on poverty for the next two decades (1980–90) and other influential concepts emerged during the same period, either in relation/reaction to Sen's entitlement concept, or independently. Power was a key concept to the debate.¹⁶ Powerlessness (or its counterpart, empowerment) refers to the means by which entitlements (access to resources) are maintained and defended. Chambers (1983, 1997) and many others (e.g., Friedmann, 1992) have stressed that the poor usually suffer from a low level of socio-political organization and that their capacity to make their voices heard is consequently weak, resulting in exclusion from

political and decision-making processes. Finally, the 1980s were characterized by wide acknowledgement of the previously neglected issue of gender-related poverty (e.g., Agarwal, 1985).

The 1990s saw the further development and specialization of approaches to the concept of poverty. The ILO's "basic needs" model was further extended by the UNDP which reutilized the idea of multidimensionality to develop its human development index approach. This time the emphasis was placed on the factors enabling (or constraining) individuals in their attempts to sustain their livelihood and pursue their development. The constraints, as defined by the UNDP (1997, p. 16), still include physical conditions such as being malnourished, being inadequately clothed and sheltered but also social exclusion (appeared first in European countries, e.g., the campaign on urban social exclusion conducted in France by Médecins du Monde (MDM) in the late 1980s). This powerlessness concept largely contributed to the rights-based approach developed by UNICEF in the early 1990s and then adopted by UNDP in its latest human development report (UNDP, 2000). Under this rights-based approach, it is suggested that the key elements of a multidimensional approach to poverty reduction should be the promotion and protection of economic, social, and cultural rights, but also civil and political rights through a (still to be developed) suitable international legal framework. Concepts such as self-esteem or self-respect were also proposed during the last decade (e.g., Beck, 1994).

To sum up, at the end of the 1990s, a greater agreement on the definition of poverty, seems

to have been achieved. In this respect the World Bank's, 2000 *World Development Report*, although it may still present some fundamental gaps, notably with respect to the rights dimension (see Maxwell, 2001, p. 146 for a discussion on this point), is representative of this "new" consensus.¹⁷ Poverty is now embodied within a multidimensional model recognizing that, in addition to low income, a wide range of factors (including social and physical isolation, lacking political voice, low social status, and health) comprise deprivation. Figure 2 provides a (nonexhaustive) list of the various concepts related to poverty. The objective of this figure is (i) to list the large number of (partially inter-related and overlapping) concepts and models that are now associated to the concept of poverty¹⁸ and (ii) to illustrate to what extent the explanatory frame proposed under the old paradigm of fisheries may be too restrictive to grasp and describe correctly the causes and nature of poverty in fisheries-dependent communities. Indeed from the literature reviewed in the first section it is clear that the way poverty was perceived and understood in fisheries lies probably quite close to the income-poverty concept at the beginning of the list.

(b) *How (and to what extent) is the new agenda of use in addressing poverty in fisheries?*

Instead of adopting without debate all the elements of the new definition of poverty in an attempt to "better" describe or understand poverty in fisheries, some questions must be raised. First, from an empirical/descriptive point of view, does this new definition "fit" with the image that we have of poor fishermen,

Time	Concepts
1900s	Income-poverty (Rowtree, 1901)
1970s	"Basic-needs" model (e.g., ILO)
	Food entitlement (Sen, 1981)
1980s	Empowerment (e.g., Chambers, 1983)
	Rights-based Approach (UNICEF)
	Gender Issues (e.g., Agarwal, 1985)
1990s	Vulnerability (e.g., Swif, 1989)
	Social exclusion (MDM)
	Self-respect (e.g., Beck, 1994)
	Environmental entitlement (e.g., Leach, Mearns & Scoones, 1995)
	Livelihood approach (e.g., UNDP, DFID, Oxfam)
2000s	Human Development (e.g., UNDP, 2000)

Figure 2. Review of (some of) the main concepts proposed in the literature with reference to poverty.

or does it modify this image? Is poverty in fisheries correctly described through this new multidimensional definition? Second, on a more conceptual/analytical point of view, how does this refined theoretical framework alter our understanding of poverty in fisheries? In other words, beyond adding a (undeniable) degree of theoretical sophistication, do these new approaches have practical value in advancing our understanding of the cause of poverty in fisheries?

Given the wide range of elements and concepts which are now part of the new consensus about poverty, it would be difficult to assert that the poorest households in fishing communities are not somehow "embodied" in this new poverty definition. Indeed, "fishing communities are often characterized by overcrowded living conditions and inadequate services, low levels of education and a lack of skills and assets (particularly land) . . ." (Townsend, 1998, p. 140). Moreover, fishers are generally "liv[ing] in remote and isolated communities, are poorly organized and politically voiceless and . . . often highly exposed to accidents and natural disasters" (FAO, 2000a, point 8). But these observations, although probably true from a broad and very general perspective, do not mean that the newly recognized aspects of poverty are specific to fishing communities *per se*. Rather they reflect the general lack of development of the rural areas within which fishing communities live. Migrant herders are probably equally, if not more, affected by these aspects of rural poverty, although this assertion is difficult to prove or dismiss since no comparative analyses have ever been made.

Therefore, although the new definition of poverty is highly relevant to illustrate and understand the complexity of poverty (and subsequently the complexity of poverty *in fisheries*), there is a danger that this new definition creates a picture of poverty somewhat too "multifaceted," and thus diverts us from focusing on the nature and cause of poverty that are directly rooted to the fisheries activities themselves.

(c) *The central role of socio-institutional mechanisms*

The view defended here is that the main lesson from the new definition of poverty, and in particular from the extension of Sen's concept of entitlement to natural resources (see for instance Leach, Mearns, & Scoones, 1995, 1999;

Mearns, Leach, & Scoones, 1998) is the recognition that there is no simple linear relationship between population, production, resource availability, and poverty. By analogy with Sen's treatment of famine, recently there has been a growing realization that to better understand poverty—as it relates to natural resources—it is necessary to redirect part of our attention and analysis effort away from the resources themselves (and their assessment) and to put greater emphasis on the role of politics of (or power over) access, control, and redistribution of these resources (see Swift, 1989; Gore, 1993; Gasper, 1993; Devereux, 1996 for further discussion on this point). Because the commands of the social actors (individual, household or group) over fishing resources depend first, on their position in their own society/community (what their ethnic group/religious affiliation is, how much land or livestock they own in addition to their fishing gear, what their social rank/kin-lineage is, etc.), and second on the institutional arrangements which legitimize and govern these commands (usually a confusing and overlapping web of local "traditional or customary" rules and *de jure* "modern" management system regulations) analyzing the cause and nature of poverty in fisheries requires the widening of the research framework beyond the economic (income level) and/or environmental (resource scarcity) dimensions. This is not to deny the fundamental and intrinsic constraint imposed by "the limited nature of the resources" (Bailey & Jentoft, 1990, p. 334) and the impact that this may have on the fishermen's livelihoods—as clearly illustrated through the example of the Kerala artisanal fisheries. However, it is also clear that a shift is needed to refocus (at least part of) the research away from the issue of natural resources limitations *per se* toward the social, cultural, and political elements which shape the relationships *between* poor people and these natural resources and between poor and less poor people.

(d) *A typology of impoverishment processes*

To justify this shift of attention, the importance of the socio-institutional dimension of poverty in fisheries is illustrated in the rest of this section through the review of some empirical studies. But, because "institutions" (and *a fortiori* "socio-institutional mechanisms") cover a rather large and vague set of mechanisms (markets are institutions, states are

institutions, local authorities are institutions, moral economy and reciprocity principles are institutions, etc.) it was found useful for this review to distinguish the different elements that may constrain or limit individuals' or groups' command over resources through a typology. This typology distinguishes four discrimination processes: social marginalization, economic exclusion, class exploitation, and political disempowerment. Although closely interdependent and sometimes reinforcing each other (as discussed below), these mechanisms are analytically separable (Table 4).

(i) *Economic exclusion*

Economic exclusion refers to leaving individuals out of a particular economic activity—in the present case fishing activity—because of their economic/financial inability to access the capital (factor of production) necessary to enter and/or operate this activity.

The concept of exclusion challenges the hypothesis of fisheries as “last resort activity” where the poorest are assumed to be able to enter the sector because of its open-access nature. Fisheries, like any other economic activities, display some entry costs/premium (purchases of fishing gear and/or license, bribes to the local government officials and/or traditional authorities who control the access to the fisheries, etc.). Further, the fact that the most profitable gears are usually the most expensive to purchase and to operate, and that the most productive fishing spots (e.g., receding channels in floodplain fisheries) are usually either “privatized” or their access requires high bribes and/or fees induces that fisheries, defined in terms of costs of entry, are far from open. In Bangladesh, for instance, Kremer (1994) analyzed the seasonal pattern of the villagers' participation in floodplain fishing in Hail Hoar (River Meghna's catchment). He observed that the poorest section of the population does not have access to fishing during the driest (receding) season when the entry premiums, which mirror the expected yields, are at their highest values. He also showed that the gears that generate a higher surplus, i.e., brush pile and pump embankment, require the most capital (including bribes and tolls). There is consequently a tendency for poorer households to be excluded from the more profitable floodplain *métiers* (association of specific fishing techniques or gear with a particular fishing ground). Kremer (1994) concluded:

Once again fishing is seen not to be a “fall-back” activity for the disadvantaged but an *activity from which the disadvantaged find themselves excluded*. One immediate cause of the poor fishing households' relatively low participation in fishing is their tendency to be excluded from the dry season fishery ... Unsurprisingly, richer fishing households catch 155% more fish by weight per household than their poorer counterparts. Since richer households are not only more likely to have dry-season access, but also more likely to be able to afford brush-piles, a much higher proportion of their catch consists of the larger, stocked species. (p. 19) ... *The poorest people are not those who fish but those excluded from the fishery.* (p. 9) (emphasis added).

This conclusion is very similar to what was observed during a recent field research carried out in sub-Saharan inland fisheries. In Chari Delta (Chad) and along the western shore of the Lake Chad (Nigeria), a comparative analysis of households' access to water-bodies was undertaken within several villages (Béné *et al.*, in press). Some of this survey's results are synthesized in Figure 3. The comparison reveals that while the richest and medium strata of the population have access to the same types of water-bodies, the poorest households only have access to a marginal part of the water-bodies exploited by the community. This difference reflects “direct” (financial) and “indirect” (technical) constraints faced by the poor. The direct restrictions result from the various legal and/or illegal taxes that are imposed for the access to the water-bodies.¹⁹ These different fees (which overlap each other) represent multiple financial barriers which affect most severely the poorest and prevent them from entering the fisheries. In addition, these poorest households also face indirect (or technical) restrictions of access resulting from their lack of adequate fishing gears and boats necessary to fish particular water-bodies (such as the open waters of the Lake). The existence of these “technical” restrictions was revealed by a comparative analysis of the sets of fishing gear owned by the different wealth groups. The survey showed that although all wealth groups use the same traditional gears, i.e., gillnets, traps (Mali traps or “*goura*”), hook-lines, cane traps (“*ndurutu*”), castnets, and dipnets (“*sakama*”), a large dissimilarity exists between the wealth groups in terms of number and size of gear. For instance, for the most common gears (gillnets, *goura*, and hook-lines) the poorest households own on average 70–90% fewer units than the richest house-

Table 4. *Typology of intrasectoral "socio-institutional" processes leading to the denial of individuals or groups' commands over the resources*

Discrimination process	Definition	Access denied	Nature of the discrimination	Comments
Economic exclusion	Process which leads to the leaving out from a particular economic activity of certain individuals due to their economic/financial inability to access the factor of production necessary to enter and/or operate this activity	Yes	Economic ^a	
Social marginalization	Process which leads to the denial of the command over a resource, service or commodities for certain actors based on social criteria such as caste, gender, or ethnic origins	Yes	Social	Like exclusion, marginalization refers to situations where actors are denied access and use of a resource (as opposed to class exploitation). The two concepts, however, differ by the nature of the barriers: economic in the former, social in the second
Class exploitation	Situation where a higher class is perceived as being in the position of extracting surplus labor from a (lower) working class, or where this "lower" class is considered as not receiving its "fair share" in the benefits created by an economic activity	No	Social	Exploitation differs from exclusion and marginalization in that it corresponds to cases where the poor are not denied access to the resource/economic activity
Political disempowerment	Situations where actors are "left out" from participation and/or decision-making processes leading to low/poor opportunities to control and govern their own commands over resources. This may result in reduction or even denying of access and use of the resources. The initial barriers are due to asymmetrical power relationships based on social stratification	Yes/No	Political (power)	Under situation of disempowerment, actors (users) may access the resources (e.g., the fisheries). What they do not access is the decision-making process (e.g., management system) which governs the modalities of access to and use of these resources

Note: This typology is not specific to fisheries and may be applied to other natural resource-dependent communities.

^a Although the nature of the "discrimination" in the case of economic exclusion is (at first sight) economic, this usually reflects either direct (market) failure, or credit facilities failures, which both can be considered as underlying institutional failures.

holds. In fact, a significant proportion of these poorest households own no fishing gear at all.

It would be tempting to conclude from these examples that access to finances and credit fa-

cilities is the central element to poverty alleviation in fisheries. Micro-credit may indeed be part of the answer. But, as the next section shows, had the poor better access to economic capital, it would not be sufficient in a large

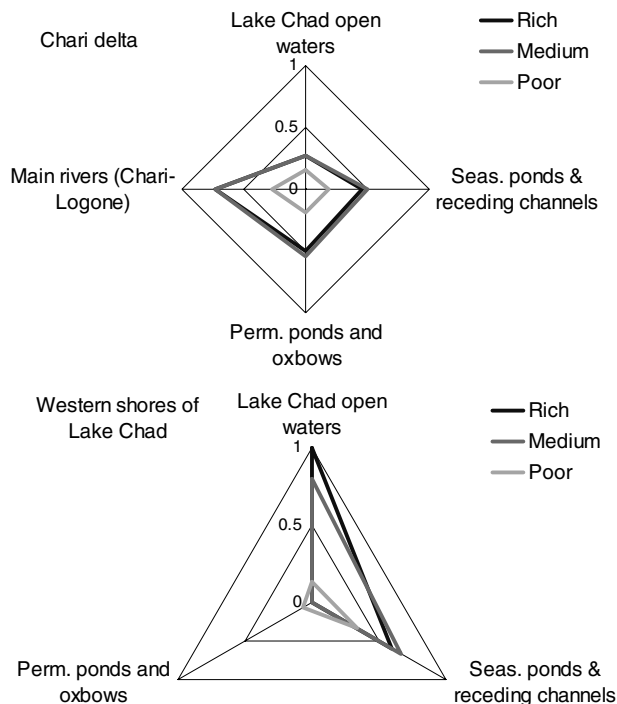


Figure 3. Comparative analysis of access to fishing grounds in villages of the Chari delta (Chad) and western shores of the Lake Chad (Nigeria). The different polygons reflect the types and proportions of water-bodies accessed by the different wealth groups (categorized as "rich", "medium" and "poor"). The accessibility ranges from 0 to 1. A value of 0 means no access to the water-body considered. A value of 1 means the water-body is accessed in 100% of the villages surveyed. Source: Modified from Béné et al. (in press).

number of cases to ensure their access to the resources.

(ii) Social marginalization

Social marginalization refers to the denial of command over a resource, service or commodity—in the present case, fishing resource—faced by certain individuals, communities, or groups of individuals based on social criteria such as caste, gender, or ethnic origin. Like economic exclusion, marginalization corresponds to situations where individuals (or groups of individuals) are denied access and use of a resource (as opposed to class exploitation—see Table 4 and next section). The two concepts, however, differ in the nature of the barriers: economic in the former, social in the second.

Kin systems, ethnic groups, and/or religious affiliations have long been recognized as playing a major role in social status and distribution of wealth endowment in rural African and

Asian societies (Agarwal, 1997; Beck & Ne-smith, 2001; Behnke, 1994; Brock & Coulibaly, 1999; Devereux, 1996; Peters, 1994; Vedeld, 2000). In fisheries Davis and Bailey (1996) recall that it is not uncommon to observe that for those considered "legitimate, rightful" participants in the fisheries to belong to specific families, kin systems, ethnic or religious groups and consequently that others (individuals or groups) may find themselves socially marginalized. Several empirical examples based on the subcategories of ethnic, religious/caste, and gender-related discrimination, are reviewed below.

Marginalization based on ethnic criteria: The access to fishing grounds of certain groups (usually those in the minority or newly settled in one area) may be denied by the local population (or even central authorities) based on ethnic consideration. In the northern part of the Yaéré floodplain (Cameroon), for instance, two major ethnic groups "share" the fishing

resources, the Kotoko and the Mousgoum. The Kotoko settled in these floodplains around the 15th century. The Mousgoum arrived later, at the beginning of 20th century. Harkes, who spend several months studying the two fishing communities, explains:

The Kotoko claimed (and still claim) ownership over the depression and the plain through ancestral rights. (...) Nowadays during the flood Mousgoum as well as Kotoko enter the plain to fish. Most techniques like cages, traps, nets fishing-hooks and even the Taraau [sort of dragnet] are accessible to [all] fishermen. (...) Two techniques though are under clear restriction of the Kotoko, the techniques that obtain the highest catches with a low labour intensity: the fishing canals and the barriers. When the barriers are used in the dry season Kotoko restriction become visible ... Mousgoum fishery then is limited to the public part of the water [i.e. the river] (Harkes, 1993, p. 39).

Thomas (1996) provides another example of restricted access in inland fisheries based on ethnic grounds. In the Hadejia-Jama' floodplains of northern Nigeria, although most of the floodplain and *fadamu* (lowest part of the dunes flooded during the wet season) is under open access or communal access (depending on the flood cycle), the sites for *sankia* traps are generally owned individually. Thomas reports (1996, p. 298) that the ownership of these sites (which are usually very productive during receding period) is largely restricted to the ethnic group native to the floodplain region (i.e., the Bede), and consequently, the other ethnic groups (Hausa from Sokoto) are denied access to this type of fishery.²⁰

Marginalization based on religious/caste criteria: Individuals or communities may also be marginalized due to their religion or their caste. In Bangladesh, for instance, Kremer (1994) reports that in the northeast part of the country the Muslim communities specialized in fishing activities (called the *maimal* or *motshojibi*, i.e., fishermen) often act collectively to prevent fishermen from other villagers from fishing over certain areas of "open-access" flooded agricultural land. They maintain their territory by physical violence and stealing gears, generally at the expense of Hindus *nomosudras* ("untouchable caste" of fisherfolk). Likewise, in Sri Lanka, Atapattu (1993, p. 570) remarks "... socio-cultural barriers play a major role in all aspects of fisheries." He illustrates his point by emphasizing that although Buddhists compose 67% of the population, they represent only 22% of the fishing population. He further asserts:

There is another factor which has some influence on entry into fisheries. Generally two castes, viz. the Karawa caste in the Sinhala community and Karayar in Tamil are engaged in the fishing industry. [And he concludes] This caste composition plays a major role in keeping the "outsiders" away ... (p. 570).²¹

Marginalization based on gender: Although the extractive part of fishing activity is usually operated by men, fish processing and trading are relatively often—especially in West Africa but also in some parts of Asia—carried out by women, either from the same household—the fisherman's wife(wives), or daughters—or from nonrelative households. These women can be subject to social marginalization (within the household or fishing communities) through gender-based segregating mechanisms. In India, for instance, Sinha and Sampath (1994) reported:

The fisherwomen (...) have a very low status in social hierarchy and social and cultural barriers normally prevent them from playing an equal part in the decision-making process even in the family (p. 153). They are excluded from membership of fisheries cooperatives. (...) It has been observed that even though fisherwomen supplement their fairly incomes through fish marketing, net making, seed collection and sales, this does not entitle them to make independent decisions or to have access to resources either within the family or in the fishing village (p. 156).

Davis and Bailey (1996, p. 259) also refer to Wahyono, Chong, and Pahlavi (1993) and Zerner (1993)—both in Indonesia—for other examples of social marginalization based on kin systems, ethnic groups, and/or religious affiliations.²²

(iii) *Class exploitation*

Exploitation derives from the Marxist concept of class struggle and refers to a situation where a higher class is perceived as being in the position of extracting surplus labor from a (lower) working class, or when this "lower" class is considered as not receiving its "fair share" in the benefits created by an economic activity. Exploitation differs from exclusion and marginalization in the sense that the former (exploitation) corresponds to cases where the poor are involved in the economic activity, whereas in the exclusion/marginalization situation, they are not (see Table 4).

In fisheries, exploitation may ensue from the extreme form of patron-client relationship. Platteau (1995, pp. 768–769) notes in this respect:

Labour relations in fishing villages are worth mentioning because they force us to reject the simplistic view according to which patron-client ties are an archaic arrangement doomed to vanish with the advent of capitalism. In fact labour tying is a pervasive feature of production relations in small-scale fishing communities and its incidence is much larger than in the agricultural sector.

The potential for exploitation or even oppression appears to be especially high when a complete asymmetry exists in the relationships between the patron and his/her client(s). Nguninguri (2000) describes, for instance, the situation of the Vili fishers of Pointe Noire (Congo-Brazzaville) where the "successful" patrons ("big men") are those who manage to tie to their own units of production the (good) fishers "who then own them gratitude and even obeisance" ²³ (p. 294). For this, the patrons develop particular relationships where they control fishers through a complex and subtle mixture of generosity-redistribution, pressure, influence, intimidation and even threats. Nguninguri describes how "the dissuasion operates through evocation of the "risks" that the fishers may face by losing the "protection" of [their patron]" ²⁴ (p. 295).

The individual patron-client relationship is not the only form of production relation that may lead to class exploitation. In Asia, the leasing system that operates on a national scale in the floodplains of Bangladesh and Mekong basin (Cambodia) is another example of an institutional arrangement relatively propitious to the establishment of exploitation relationships between social classes. Under this leasing system, the water-bodies, which are the property of the government, are compartmentalized into individual "fishing lots": *jalmohals* (Bangladesh) or *loh nesaat* (Cambodia). These water estates are then leased, typically for one to three years, through an auctioning system—which is thought to yield very substantial revenues to the government. But, because the lease fees represent large amounts of money, fishermen themselves (either individually or even through co-operatives) can rarely raise the necessary funds. Consequently the leaseholders are usually rich investors/entrepreneurs, typically from the rural elites. Toufique (1997) for instance (p. 459) explains:

Most of the lessees are past or present members or chairmen of local institutions of the state. In the rural context it is the most influent and powerful people who compete for such posts. So the lessees usually

come from more powerful section of the rural population

These lessees—also called "water-lords" ²⁵—hire fishermen as daily labor to carry out fishing operations on their behalf. Daily laborers are paid in cash and/or receive a small share of the catch while most of the sale values accrue to the leaseholders. An abundant literature describes in detail the ways these leasing systems have led to the institutionalized exploitation of the fishermen by the rural elite class. See, for instance, Thompson and Hossain (1998) or Toufique (1997) for Bangladesh or Degen, van Acker, van Zalinge, Thuok, and Vuthy (2000) or van Zalinge, Thuok, & Seang Tana (1998) for Cambodia. In Bangladesh, Bernacsek (1994) emphasizes (pp. 3–5):

The Jalmohal lease system is oriented towards generating revenue for government and substantial income for leaseholders. However, *it exacerbates exploitations of fishermen* (who do not own their own produce and therefore *realise only a small part of this full economic benefits*) ... the share of profits yielded to labour ... is minimized to the extent possible, within the constraints of labour's opportunity costs and its awareness of fishing as a profit rich economic activity (emphasis added).

Likewise in Cambodia, Degen and Thuok (1998) conclude (p. 13):

In general the relation between the villagers [fishermen] and the lot owners is characterised by tensions, threatening, conflicts, commercial partnership of sub-leasing and partnerships in fishing operation. ... The weakness of the given [leasing] system is not that it does not work or might be inefficient, but rather that it is as arbitrary as in ancient times [i.e. during the French "Protectorate"] and allows for *abuses on the stronger side of the relationship* (emphasis added)

(iv) Political disempowerment

Disempowerment is a concept slightly more difficult to define. Friedmann (1992, 1996), for instance, recognizes three dimensions to disempowerment: social disempowerment, referring to "poor people's relative lack of access to the resource essential for the self-production of their livelihood"—a somewhat large concept which embraces the three previous categories identified above; a psychological disempowerment, referring to "poor people's internalized sense of worthlessness and passive submission to authority"—relatively close to the issue of self-esteem or self-respect proposed by Beck

(1994); and a political disempowerment referring to "poor people's lack of clear political agenda and voice" (Friedmann, 1996, p. 165).

The conception of disempowerment adopted here is relatively close to the third dimension of Friedmann's disempowerment, i.e., the political dimension. In this essay, disempowerment refers to situations where the poor are "left out" of political participation and/or economic decision-making processes. Within this approach the problem of power is primarily seen as one of (inequality of) opportunities of people's about their own commands. "If ... one party was excluded in the deliberations [on commands over resources], or if that party was ignored or overridden, then power has been exercised" (Bromley, 1989, p. 871). As recognized by Vedeld (1998, p. 8) "Power is particularly employed by stronger groups against weaker." To the extent to which elites (higher social class, traditional authority leaders, or State officials) are not just products but also makers of institutions (including markets and traditional management systems of natural resources), these elites will try to set or alter these institutions to legitimize their decisions and behavior and sustain their own initial social, economic or political advantages. The poor are usually those who are affected by these processes.

The exclusion of the poor based on power can take different forms which are usually context-specific. It is therefore difficult to cover the whole range of circumstances in which discrimination against the poorest (disempowered) takes place. It is possible however to give some examples to illustrate this in the case of fisheries.

Johnsen (1992), through his description of the fishing community in Chiloane Island (Mozambique), provides a straightforward example. He shows how the better-off households of the Chiloane community exert their power to alter market mechanisms to their own advantage. On this island, the majority of the population makes a living from collecting and drying the by-catch from the shrimp trawlers which operate off-shore. Over the past few years some families have taken a dominant position in the purchase of by-catch. Paying the owners of trawlers a considerable sum in advance, they get priority to buy the by-catch. In parallel, small-scale and individual by-catch driers, who purchase the fish from crew-members of the trawlers—who supplement their own wages through this sale—, pay instantly in cash with

small funds. Not surprisingly, due to the power asymmetry in the market transaction, Johnsen observes that "[the poorer, small-scale, driers] frequently see themselves deprived of fish especially in a situation with small by-catches." Note in passing that the crew-members are also denied the benefits normally derived from the transaction with the small-scale driers. Johnsen admits that if:

... it does not seem likely that large-scale driers, at least as long as ownership relations are separated, shall completely take over the by-catch purchases ... the balance of power between the owner/large-scale drier on the one hand, and the trawler crew/small-scale driers on the other, is critical (1992, p. 123–124).

Critical as well is the asymmetrical power relation on which fishing-lot lessees and local elites in Cambodia rely to deny the local population access to the so-called areas set aside for the people. These specific parts of the fishing lots are theoretically, i.e., by law, recognized common property areas during the open season to allow local villagers to fish for subsistence (with some restriction on gears). In practice, however, the access to these common property areas is frequently controlled by the local elites of the nearby villages or the lessees of the fishing lot within which these common property areas are located. Swift (1997) reports, for instance, that in Takeo province the local authorities lease them out to villagers in return for a fee. In other instances, fishing-lot operators prevent the villagers from gaining access to these common property areas by maintaining their armed guards in the lot area during the open season. Degen *et al.* (2000, pp. 18–19) even report:

... in Kompong Thom province, the operator of fishing lot No.6 closely co-operates with a military division based inside the lot area. The military protects the fishing lot area the whole year in exchange for exclusive rights to fish trading in certain parts of the fishing lots ... What are supposedly common property areas are frequently appropriated and leased out by village or commune chiefs or army commanders.

This process of "enclosure of the commons" through power-based relationships has a great effect on those who cannot afford the leasing fees (see section above on economic exclusion) and rely mostly on these "areas set aside for the people" to sustain their livelihoods. This section of the population is usually the poorest, that is, economically excluded and politically

disempowered. These mechanisms are similar to what is described elsewhere for land-based resources—see, for instance, Jodha (1985) or Beck and Ghosh (2000).

One more example of disempowerment: in Bangladesh, over the last two decades, a large number of fisheries enhancement programs (FEPs) have been implemented, with a double objective: ecological, as a mitigation measure for losses to fisheries due to flood control and overexploitation; social, as a way to improve the living of the poorest (landless) households which largely dependent on fishing activities to sustain their livelihoods. The technical principle of these FEPs is the release of massive quantities of fingerlings in water-bodies in the hope that the artificial introduction of young fish will result in larger adult populations and subsequent catch levels. A majority of these FEPs have been technically successful and there is little discussion about the impact of increases in fish production (see, for instance, Ali & Islam, 1998; Cowx, 1999; Welcomme & Bartley, 1998).

There is, however, increasing evidence that these FEPs have partially or even totally failed to improve the food security and welfare of the poorest people living in the nearby areas. A review of the literature (e.g., Ahmad, Bland, Price, & Kershaw, 1998; Ali & Islam, 1998; Apu & Middendrop, 1998) reveals that mechanisms such as those of economic exclusion and class exploitation have played a major role in the redistributive failure of FEPs. In brief, the constraint or lack of access to formal credit for poor fishers was ignored and no alternative to the service provided by moneylenders was offered. Consequently, the distribution of benefits derived from these new FEPs merely reflects the patterns of original ownership of the means of production, leading to (either or both) economic exclusion and class exploitation. The poorest, who could not invest in adequate fishing gear and fishing licenses (required to access the newly enhanced water-bodies) were totally excluded or only benefited from a very limited portion of the increased fish production (Bernacsek, 1994; Capistrano, Ahmed, & Hossain, 1994).

However, a more careful reading of the literature reveals that underneath these evident mechanisms, a more subtle effect resulting from the disempowerment of the poorest was also at work. As human inputs increased through FEPs, the control of the aquatic environment and the access to the floodplain resources tended to be transferred from the public/com-

mon to the private domain (a sort of enclosure), so that those investing in the FEP activities could protect their future interests. This change induced an assignment of ownership or use-rights to individuals or groups of individuals of what was previously an open-access resource. This transfer of rights mirrored the local power structure of the Bangladeshi rural society and the poorest (i.e., most disempowered) were usually left out from this “redistributional” process. Thompson and Hossain (1998; quoting BRAC, 1995), for instance, report that in the case of the FEP Oxbow Lake Project II, 44% of the local population whose access was denied to the new enhanced fishery were initially poor and landless households deriving the main part of their livelihoods from the different *boars* (oxbow lakes) “enclosed” in the project. Tactfully, Thompson and Hossain (1998) conclude “Hence for various reasons (sic) there was some inequity in the extent of participation and benefits from the project” (p. 360). Ahmad *et al.* (1998) are more direct in their conclusion. With reference to the FEP Third Fisheries Programme (TFP), they assert:

Power structure and fishing patterns that have been in place for many years cannot be changed overnight. Indeed, increasing the value of the resource [through FEP] can lead to poor fishers being forced out of their usual fishing grounds by more influential groups in the community. This was shown in BSKB [one of the *beels* (floodplain pool) included in the TFP] where support from an NGO, Prodipan, was necessary to ensure that poor fishermen gained access to the enhanced fisheries resource ... (1998, p. 343).

In fact, in total, no less than 18 nongovernmental organizations (NGOs) had to be involved during the last year of the Third Fisheries Programme, partially to ensure that poor were not further excluded.

These different observations suggest that FEPs might have *de facto* led to an increase in inequity through the transfer of access and benefits of the common resources from the poor landless to the rich rural elite. Although part of the unequal (re)appropriation of FEP benefits resulted from the pre-existing economic and social (class) inequities characterizing rural Bangladeshi society, unbalanced power relationships further aggravated this phenomenon through exclusion of the poorest from the different stages of the decision-making process and implementation of the FEPs.

(e) *The need for a new agenda*

This last example is particularly instructive for several reasons. First, it clearly illustrates that poverty in the fisheries sector may still occur even in the absence of Malthusian crisis (even more, in the present case, where the increase in production and catch of fish following the enhancement has been clearly established in several FEPs). It also suggests that there is no one-to-one relation between fishery and poverty, and more importantly, that the redistributive dimension which is almost systematically assumed to take place in fisheries (as an open-access activity), is not in fact always the day-to-day reality of the fishing poor.

Second, disaggregating the poor into (socially) marginalized, (economically) excluded, (politically) disempowered and (class)-exploited groups reveals a range of different mechanisms working through an intrasectoral (fisheries) dynamics and leading to the impoverishment of a part of the community. What the FEP example suggests, however, is that these different mechanisms of exclusion, although (at least on paper) distinguishable, are in reality often associated, overlapping, or even interlocking. Thus, it is relatively difficult in the case of the Bangladeshi rural society to make the distinction between disempowerment and class exploitation, since the exploiting (economic) class (i.e., the rural elites) is also the one pulling the strings of local political power. Likewise, whether the disempowerment status of the poor in these floodplains is the cause or the consequence of their economic exclusion is a rather difficult question which deserves further research and analysis.

It seems therefore that the categories used in the typology do not reflect perfectly the complexity of the mechanisms at work. In his defense the author would restate that this classification is only a typology and as such is merely descriptive. In particular, it does not offer any analytical handle to grasp the underlying motivations or origins of the segregation mechanisms²⁶. There is here a very large domain of research which unfortunately has been largely neglected (as far as fisheries are concerned). A number of questions both in terms of research and policy implications, come to mind, for instance, on the degree of equity that is usually assumed to characterize the indigenous institutions in charge of the resource management (the so-called traditional management systems); on the magnitude, un-

derlying mechanisms and redistributive consequences of the process of privatization (enclosure) of the inland fisheries—currently taking place at an increasing rate both in Africa and Asia; on the implications of the overlapping of traditional—or neotraditional—and State authorities on the redistribution (re-appropriation?) of the benefits generated by the fisheries; and on the consequences that this overlapping induces for the poorest (i.e., disempowered). These are some of the fundamental questions that must be raised if one wishes to address the real issues of poverty and poverty alleviation in small-scale fisheries in Africa and Asia.

5. CONCLUSION

This article was originally motivated by the absence of references to fishers and fishery communities in the research currently carried out on poverty, despite the high degree of poverty reported in fisheries literature for more than 50 years.

A closer look at this literature revealed that the “debate” on poverty in fisheries has been reduced to an almost universally accepted perception that “fisheries rhymes with poverty” and that the roots of the problem were the economic and biological aspects of the activity. Poverty in fisheries was explained through a linear relationship between the low incomes of fishermen (due to low catch) and the overexploited resources (created and/or maintained by the open-access nature of the fisheries). Although it is emphasized in the first part of this article that it is important to recognize the intrinsic biological limits of the resources and the undisputable negative impacts that resource degradation or depletion have on fishermen’s livelihood (both through income but also food supply), the article showed that the focus on the supply side of the activity and its underlying Malthusian “tragedy” may have generated a circular and autoreinforcing paradigm (based on two pillars “they are poor because they are fishermen” and “they are fishermen because they are poor”). This dynamic prevented scholars and analysts from developing an adequate framework necessary to emulate a fruitful debate about poverty in fisheries. FAO, for instance, admits: “Currently, the normative areas of work by the Fisheries Department (FI) do not explicitly address poverty in fisheries” (FAO, 2000a, point 3).²⁷

It seems that this lack of an adequate analytical framework may also explain the current "poverty of the contribution of fisheries on poverty issues" and is probably one of the main reasons why analysts from other disciplines do not use fisheries-related examples in their current research. Over the last three decades, in these other disciplines, dramatic improvements have been made in the comprehension of the nature and cause of (rural and urban) poverty. One of the major conclusions of these recent research efforts has been that poverty is a complex phenomenon which encompasses, alongside low income, other concepts such as illness and lack of education, social exclusion, entitlement failure, vulnerability to shocks and political powerlessness. Although all these different "facets" of poverty are important elements for the improved understanding of poverty, of particular relevance to the specific debate of poverty in fisheries is the conclusion initiated by Sen's work and recently extended by others (e.g., Leach, Mearns, Scoones, etc.). These authors have pointed out that poverty affecting different social actors (individuals, households, or communities) does not systematically reflect a Malthusian crisis, but may be related to an entitlement breakdown, i.e., a failure for these actors to access (natural) resources, food, commodities, and services.

Applied to fisheries, this finding has helped us to reconsider the conclusions distilled for decades by the old paradigm and in particular to recognize that the different socio-institutional mechanisms which govern the command of fishery resources (essentially social positions and the institutional arrangements controlling the access to, and the use of, these resources) play a more critical role in deter-

mining poverty (or symmetrically wealth) than pure economic or biological considerations. Following this, a very simple typology was then proposed in the second part of this article, which highlighted four different categories of socio-institutional mechanisms leading to the denial of individuals or groups' command over the resources. These mechanisms of exclusion were then illustrated through empirical studies derived from the fisheries literature.

Through the analysis of these case-studies, it is suggested that (a) although poverty is now recognized as a multifaceted issue—requiring therefore transsectoral assessment and policies approaches—there are specific intrasectoral mechanisms which can lead to, or maintain, the impoverishment of segments of the fishing community; and (b) that the interest that fisheries should receive in relation to poverty (both from inside the fishery discipline and from nonacademics and policy-makers) is not related to the production specificities of the activity (in particular the limited nature of fisheries' resources) but rather to the institutional "richness" which characterizes the social relations between the different actors involved in this activity. Fishing communities represent an "unexhaustible mine" of examples of collective action, decision-making processes, power relationships, social (re)distribution or (re)appropriation mechanisms, and local political competition (for the access and control of the resources and rents generated by these resources) which constitute some of the central themes of research currently addressed through the new agenda on poverty alleviation. Small-scale fisheries should therefore be at the core of this research.

NOTES

1. One occurrence on p. 68 in the Box 4.3 "Agricultural reforms in Chile and China help small-farmer" ... and two occurrences on p. 176 in the sentence: "They [the workfare programs] can also help people affected by less visible impacts of a disaster, such as the poor *fishers* in Ecuador and Peru who fell deeper into poverty as *fish* fled the waters warmed by El Nino" (emphasis added). For comparison the key words "farm" (including farm, farmers, farming, nonfarm, farmland, etc.) and "forest" (including forest, deforestation, reforestation, forest management, etc.) appear respectively 38 and 14 times more frequently (respectively 115 and 43 occurrences).
2. A similar exercise conducted on the Rural Poverty Report 2001 (IFAD, 2001) generated the following outcome: "fish": 32 occurrences, "farm" 1500+ occurrences, "forest": 82 occurrences.
3. As admitted by some of the poverty analysts, fisheries are very much outside their areas of expertise (e.g., Mick Moore, Institute of Development Studies, Brighton, electronic communication, August 3, 2001).
4. This perception of a strong relationship between poverty and fisheries has been challenged explicitly only

in very few documents. One of them is Bailey's (1994) FAO paper, where the author stated "The notion that small-scale fishers are the poorest of the poor can be challenged on logical grounds by observing migration into rather than out of many fishing communities" (p. 27). Note, however, that Bailey then provides evidence of this immigration process (p. 29) but not that fishermen are actually less poor than nonfishermen. Furthermore, it could be argued that the poverty outside the fisheries will eventually induce poverty in the fishery sector through labor market re-equilibrium (see next section for a discussion on this point).

5. For instance, few months ago one could read in the ODI Natural Resource Perspectives "Both marine and inland artisanal fisheries tend to include the poorest sectors" (Payne, 2000, p. 1).

6. This anecdotic point was brought out by Wright (1990, p. 1) although already mentioned by Clark (1985, p. 1).

7. See Ciriacy-Wantrup and Bishop (1975), Berkes, Feeny, McCay, and Acheson (1989), Gibbs and Browley (1989) among others about this issue.

8. Both Gordon and Hardin refer to this Malthusian mechanism in their respective paper—(Gordon, 1954, pp. 125–126) and Hardin (1968, p. 243).

9. Note that one implicit assumption in this approach is that people cannot leave the fishery once they enter it. This assumption of "nonreturn valve" may partially reflect the reality in developed countries' where fisheries are characterized by high sunk costs induced by large investments and capital-intensive technologies. This assumption however is more disputable in the case of small-scale fisheries (especially in developing countries) where the levels of investment are usually lower (R. Willmann, personal communication, November 15, 2001).

10. Note, however, that the relation between the decrease in catch volume (following the depletion of the stock) and the impact on the fishermen's incomes is not necessarily so simple. As Bavinck (1996) in the case of the artisanal fishery along the Coromandel coast (eastern India) reports: "As a result of extensive trawling and possibly other influences, fish stocks have declined to dangerous levels. Declining catches have been compensated by steadily rising prices, however, so that the incomes of fishers have not been affected in a serious way" (pp. 476–477).

11. Perhaps the most famous example in this respect is the collapse of the Peruvian anchovy fisheries in 1971–72 (Paulik, 1981) where the sharp decline in the catch

induced dramatic dislocations in the whole Peruvian fishing industry: crewmen were released by boatowners while boatowners themselves were going bankrupt, factory workers were released by owners of processing plants, while processing plants were shut down, etc.

12. Perhaps Copes's (1989) discussion paper "Why are fishing incomes often low" should also be mentioned here. Although he used a slightly different argument than Cunningham, his main conclusion rejoins Cunningham's, namely that fishermen's incomes mainly reflect the low opportunity costs that characterize small-scale developing countries fisheries.

13. The formal demonstration can be found in Anderson (1980, pp. 861–862).

14. In this respect, it is clear that the "income poverty" approach which has dominated the research on poverty assessment until the late 1970s has obviously greatly influenced the perception that academics and decision-makers have developed during the same period about poverty in fisheries. For a further discussion on the limit of income poverty, see the second part of this article.

15. This definition of poverty based on income or consumption (the so-called income poverty approach) originated from the early works of Rowtree (1901) in the suburbs of London.

16. Vulnerability is another concept which has been recognized as strongly related to (although distinguishable from) poverty. It refers to exposure to shocks and stress, and the ability to cope with them. This emphasis on vulnerability originates from works by Chambers (1989) and Swift (1989)—see also more recently Watts and Bohle (1993)—and highlights the temporal dimension of poverty, particularly making the distinction between chronic and transient poverty. This perspective also emphasizes the importance of assets as buffers (moral economy, social capital) and safety nets.

17. This overview of international new "consensus" on the question of "What is poverty and how can we define it?" would not be completed without mentioning the major methodological change that took place in the 1990s through the development of participatory poverty assessment techniques. These techniques (which can be seen as another direct consequence of the empowerment concept) recognize the importance of involving the poor themselves in the data collection and assessment process regarding the extent, cause, and nature of their "own" poverty (Norton, Aryeetey, Korboe, & Dodge, 1995). It is now increasingly acknowledged that these methods, which confirm the broad (multi-)dimension of

poverty, should be part of any poverty assessment program and in that respect should be used in combination with the more conventional (quantitative household survey) approaches.

18. Baulch (1996) in his editorial about the "New poverty agenda," proposes a "pyramid of poverty concepts" (see his Figure 1, p. 2).

19. The detailed analysis of the local institutional arrangements in these areas (Béné *et al.*, 2003) shows that in 100% of the villages surveyed in the Chari delta and along the western shores of the Lake, acquiring the rights of access to a restricted fishing ground involves some form of fee payment, either in cash or as a proportion of the catch (or both). A large part of these fees are levied by the local traditional authorities, but the survey also reveals the existence of large-scale illegal taxation systems operated by soldiers of the Joint Patrol Forces (international armed forces from the four riparian countries) or even by central government agents.

20. It could be argued that this issue of exclusion/marginalization based on ethnic criteria is slightly more complex than framed in this article. Although it is presented here as a negative outcome from a poverty and exclusion perspective, in contrast it is positively appraised in a certain literature (e.g., community-based natural resource management) where it is seen as a possible "weapon of the weak" to ensure the protection and conservation of traditional community patrimony from the exploitation or expropriation by more powerful state or "capitalist" actors.

21. What is not clear, from this article is whether these castes have developed tangible mechanisms to concretely prevent other groups from entering the fisheries or whether this differential in composition merely reflects a historico-cultural differentiation between agricultural and fishing activities (the Buddhists specializing in agricultural/land-related activities and the two castes mentioned specializing in fishing activities, leading to some sort of "fishermen identities"). In fact, in this case, a review of other related articles (Atapattu & Dayaratne, 1993; Atapattu, 1994) reveals no substantial religious and/or ethnic issues and therefore suggests the second hypothesis, i.e. a historical-dependent specialization.

22. As far as marginalization based on gender is concerned, one anonymous referee noted that the gender-based division of labor in fishing communities may also work to women's advantage—that fishermen's wives may enjoy greater (relative) access to and control of family income, and greater "voice" in household

decision-making, than their counterparts in other rural communities. This is indeed a question that deserves some attention. Unfortunately, to the author's knowledge, no such comparison between fishing-household women and nonfishing household women has been proposed in the literature.

23. Translated by the author, the original quote is "... parmi les bons pêcheurs qui lui doivent reconnaissance et même obéissance".

24. Translated by the author, the original quote is "la dissuasion s'opère en évoquant les risques auxquels s'exposent les pêcheurs en perdant la protection de [leur patron]". Note, however, as emphasized by one anonymous referee, that in some circumstances the division between capital and labor may not systematically work against the "little men." Just as sharecropping is not systematically an exploitative relationship, the patron-client relationship between equipment owners and small-scale fishers is not always disadvantageous to the fishers. Much depends on the "goodwill" of the patrons and the nature of the (often traditional) institutions that govern such relationships.

25. This term "water-lords" is used by Huq and Huq (1985, p. 84), Rahman (1987; quoted in Toufique, 1997, p. 458), Kremer (1992, p. 1), Ahmed, Capistrano, and Hossain (1997, p. 237) among others.

26. Some of these motivations, however, may "percolate." For instance, in the case of the Mousgoum in the Yaéréu floodplain or the Hausa in the Hadejia-Jama'are *fadama*, it is clear that the social marginalization that these groups are experiencing from the Sokoto and Bele respectively is, in fact, used by the latter as an excuse to keep the former away from huge economic benefits generated by the most productive fishing techniques. This interpretation could, however, be perceived as a caricature by those who would criticize it on the basis that it misses the subtleness of the socio-ethnological mechanisms at work.

27. The situation may change, however, in the near future. In 1999, the Sustainable Fisheries Livelihoods Programme (SFLP) financed by the UK Department for International Development (DFID) and implemented by FAO was launched in 25 West African countries with the explicit objective of eliminating poverty in fishing communities (FAO, 2000b). Within this program, research is currently carried out on how the Sustainable Livelihood Approach (SLA) developed among others by Scoones (1998) at the Institute of Development Studies (University of Sussex) and later adopted by DFID may be used as a tool for assessing and addressing poverty in

the specific context of fisheries. Attempts to make use of other assessment tools such as poverty profiling techniques (Pittaluga, in press) or Food Insecurity and Vulnerability Information and Mapping Systems (FIV-IMS) are also being made. Within the fisheries department at FAO, several analysts through the Advisory

Committee on Fisheries Research, are also currently working on the issue of poverty in fisheries especially with reference to the Code of Conduct for Responsible Fisheries. A Workshop "Joint Working Party on Poverty in Small-Scale Fisheries" was held in Rome on April 10–12, 2002 in this respect.

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